Sahaj Bhadja

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Summary

Security-curious software engineer with **C++ fundamentals, operating systems, concurrency, performance optimization, and cross-platform Windows, macOS, and Linux delivery**, plus experience building fault-tolerant, observable production components.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Built reliable Python services with explicit state transitions, failure isolation, and input validation, processing 5,000+ monthly jobs while increasing concurrent throughput 3x and sustaining 99%+ production reliability.
- Added observability through structured error handling, job-state tracking, validation, and reconciliation, diagnosing performance issues and reducing database latency from 1.2 seconds to 350 milliseconds for 500+ users.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Implemented multithreaded processing using ThreadPoolExecutor and secure file ingestion, handling 1,000+ user-supplied PDF, DOCX, ZIP, text, and image inputs per batch with validated, fault-tolerant execution at scale.
- Investigated OCR failures, model hallucinations, timeouts, and API instability, implementing fallback behavior, diagnostic logs, and automated tests while maintaining stable performance across 10,000+ daily requests at scale.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Developed performance-sensitive Python and PyTorch components processing 100,000+ trajectory records, improving long-horizon planning 18% through architecture analysis, simulation, debugging, reproducible testing, and iterative feature integration at scale.
- Modeled interactions across ego, road, and nearby-agent signals, improving closed-loop stability 15% while reliably evaluating edge cases, tracking system behavior, and documenting technical decisions for senior researchers.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Developed cross-platform enterprise software across Windows, macOS, and Linux, automating installation, executable builds, and 20+ unit and integration tests while increasing nightly pipeline efficiency 68% at scale.
- Resolved defects in Java and Python components using object-oriented patterns, caching, and field filtering, improving response times 30%, reducing payload sizes 25%, and lowering technical debt 20%.

Cyber Security and Data Mining: Detecting Malicious URL **India**
Data Analyst - Research Fellowship *May 2023 – August 2023*

- Built a security detection pipeline over 650,000 URLs, engineering 21 behavioral indicators for IP presence, suspicious keywords, and structural anomalies to identify phishing and malware with 96%+ accuracy.
- Evaluated adversarial URL behavior using Random Forest, recurrent, and convolutional models, reliably comparing detection tradeoffs and generating actionable security insights from high-volume malicious and benign input patterns.

Skills

Systems: C++, C, Python, Linux, Unix, Windows, macOS, operating systems, scheduling
Performance: Concurrency, multithreading, resource allocation, caching, profiling, fault tolerance
Security: Security analytics, threat detection, malicious inputs, access controls, authentication, firewalls
Quality: Unit testing, integration testing, debugging, observability, monitoring, logging, code review
Engineering: Git, CI/CD, Jenkins, Docker, OOP, data structures, algorithms, system design, Agile

Projects

CPU-scheduling Simulator Live | *C++, OS Fundamentals*

- Implemented six C++ scheduling algorithms with edge-case handling, automated waiting and turnaround metrics, and Gantt visualization, testing 1,000+ process combinations while reliably reducing average turnaround time 25%.

Chat2SQL Code | *Cross-Platform Enterprise Software*

- Developed and debugged Java and TypeScript components supporting 10+ databases, configuring Docker-based workflows across Windows, macOS, and Linux while reliably implementing schema synchronization and structured data access.

FinNexus Live | *Data Security, Real-Time Systems*

- Built a security-focused platform supporting 50,000+ users and one million transactions with token-based firewall controls, secure Plaid APIs, WebSocket updates, Kubernetes deployment, and reliable, scalable MongoDB access.